

BUILT — Circuit Lab, deepened (v0.44.0)

Active devices, save / load / export, and parts that link to the catalog — on the same validated MNA engine. Still an overlay on the real schematic.

1 · NEW ACTIVE DEVICES (all unit-tested in the engine)

AC source

time-varying sine $V(t)=A\cdot\sin(2\pi ft)$; drives real transient/ringing demos

MOSFET (N-ch)

square-law level-1, Newton-solved; switches & amplifies (on: drain→0.02 V)

Op-amp

ideal high-gain VCVS; non-inverting ×2 gives exactly 2.000 V

Relay

behavioral coil + contact; 12 V energizes the 120 Ω coil → contact CLOSES → lamp lights

Relays matter for vehicle electrical systems — coils and contacts are everywhere in the wiring TMs.

2 · WHAT MADE IT POSSIBLE + THE NEW WORKFLOW

Generalised N-pin model

- Parts now carry a pin-offset table (not just 2 leads)
- rotation rotates the offsets; union-find still assigns nodes
- 3-pin (MOSFET, op-amp) and 4-pin (relay) devices just work
- netlist() emits each device in the engine's pin order

8 demo circuits

- divider · RC · RLC · diode+LED
- AC+RC · MOSFET switch · relay-driven lamp · op-amp ×2
- each validated end-to-end (build→netlist→solve)

Save / load / export

-  Save /  Load — persists in the browser (localStorage)
-  .json — download the full circuit;  import it back
-  netlist — SPICE-style .cir (R/C/L/V/SIN/D/M/relay notes)
- portable: share a circuit as a file, reopen it anywhere

Parts link to the catalog

Tag any component with a TM part # / NSN. It shows on the symbol and is a one-click jump to the Look-Alike Parts recognizer (/partdiff) — so a drawn relay or resistor connects straight to the real cataloged part and its variants.

GROUNDED & ADDITIVE (R1/R6) · RPS (R7)

Same validated MNA core (all 10 unit tests pass: divider/RC/diode/RLC + AC/MOSFET/op-amp/relay). The circuit is grounded in what you build; the simulator never rewrites the TM and the cited sheet stays behind the overlay. Save/export are local files — nothing leaves the machine. RPS: still a modern-browser feature; the Win7/Vista legacy build keeps the static-overlay fallback. Fully additive — same routes, richer editor.